A one-parameter crack detector on SEM micrographs, and the pixel budget it spends

DRAFT — not submitted. No author, affiliation or venue is set, and there is no reference list yet; both are required before this goes anywhere. Most numbers are generated from measurement files by paper/make_paper.py, which fails the build on an unresolved name, but a residue in Sections 2, 4, 5 and 6 is still typed by hand and is being migrated.

Abstract

We describe a crack segmentation detector with a single free parameter and no learned weights, and measure it on the only public crack data we could find in its own modality and material class: 31 whole scanning-electron fields of Inconel 718 with pixel-exact masks.

We previously reported that its one constant did not need refitting. That was an artefact of where the sweep stopped. The published sweep tested seven values of mad_k, the largest of which was the shipped 5.0, and F1 was still rising when the grid ran out. Extending it to 40.0 puts the optimum at 6.0, and the gap is not noise: on the whole fields 6.0 beats 5.0 in 27 of 31 images (paired Wilcoxon $p < 0.0001$) and improves the area fraction from 1.65x the labelled value to 1.27x in 23 of 31 ($p = 0.0124$). The constant is one rung off, in the direction of marking less. We report this because the claim it replaces was this paper's headline.

We report the accuracy primarily as a pixel budget rather than as F1 or IoU. On an image that is 1.80% crack, F1 and IoU discard the 98% of pixels that are correctly left alone, and they cannot state the quantity this tool actually outputs — an area fraction. The detector marks 2.34% of the image where the human marked 1.80%: of that, 0.78% is right and 1.56% is wrong, so an area fraction read off this tool is 1.30 times too high. Accuracy is 97.42%, which is worse than marking nothing; balanced accuracy is 69.82% and MCC 0.3402. In the conventional units it is F1 0.3444 and IoU 0.2137, 9.8 times the F1 of marking every pixel.

Three results cut against the method and we report them rather than omit them. Against six classical detectors tuned identically it ranks fifth of seven on these whole fields (0.3444 against the leader's 0.4697), and third of seven on the dataset's released crops, within 0.0186 F1 of the leader there. Cropping these same images around their cracks — as the dataset's own test split does — moves the reported area fraction from 1.30x to 0.78x and F1 from 0.3444 to 0.4958, with no change to the detector at all. And the ranking itself is buyable: dilating this detector's own output, which reads nothing new from the image, takes it to first place on the crops at the cost of a 2.54x area-fraction error (Section 3.5).

Status of this draft. This is one alloy and 31 images. Section 7 says what that is and is not enough for.

1 Why another crack detector

There are many. Most are learned, most report F1 or IoU against a benchmark, and most are compared against each other at their own authors' operating points. Our interest is narrow and practical: a materials laboratory needs crack measurements out of scanning-electron and transmission X-ray micrographs, on a machine with no GPU, from a tool a researcher can audit line by line and correct by hand. That rules out most of the field, and it makes the natural baseline not a network but a threshold.

The detector here is close to that baseline. It flattens the background at seven scales,

takes pixels that are dark at any of them, applies a robust outlier cut and an absolute darkness floor, and cleans the result. There is no training step and no model file.

On "one parameter". One constant, mad_k, is exposed and swept; that is the sense in which the title means it, and it is the only one given a grid in any comparison here. It is not the only constant. absolute_dark_thresh = 10 is a second gate, held fixed rather than fitted in this paper but chosen by leave-one-specimen-out on our own corpus, where it supplies roughly half the proposal-stage recall — on the alloy measured here it supplies almost none (Section 3.6), and three instruments in our own records disagree about its value. Behind those sit a seven-rung scale ladder and two size floors, frozen. A reader comparing this against a single-parameter classical filter should know that the filter carries one constant and this detector carries a frozen preprocessing budget.

Related work. Reporting a segmentation as an area budget rather than as an overlap score is not new: it is standard practice in land-cover accuracy assessment, where Olofsson et al. [6] set out the error matrix in units of area proportion together with a bias-adjusted area estimator and its confidence interval. Materials microscopy has the same instinct — Durmaz et al. [7] report the downstream morphological quantity beside the segmentation score — and a materials laboratory's own standard, ASTM E562 [8], is written in area fraction and in nothing else. This paper claims no novelty for that framing. What it offers is the measurement: the full budget for one detector on one alloy, in the units the laboratory reports, together with the ways the usual scores mislead about it. The classical arms it is compared against are the standard ones [1–5].

Reporting such a thing honestly turns out to be harder than building it, and most of this paper is about the reporting. The detector is a threshold; what is less obvious is that the scores conventionally used to grade one — F1 and IoU — cannot express the error a materials laboratory actually suffers, which is a crack area fraction that comes out too high or too low by some factor. Section 3.3 reports that factor, and Appendix A gives it for every set we measured. It ranges from 0.74x to 76.65x for the same unchanged detector.

2 The detector

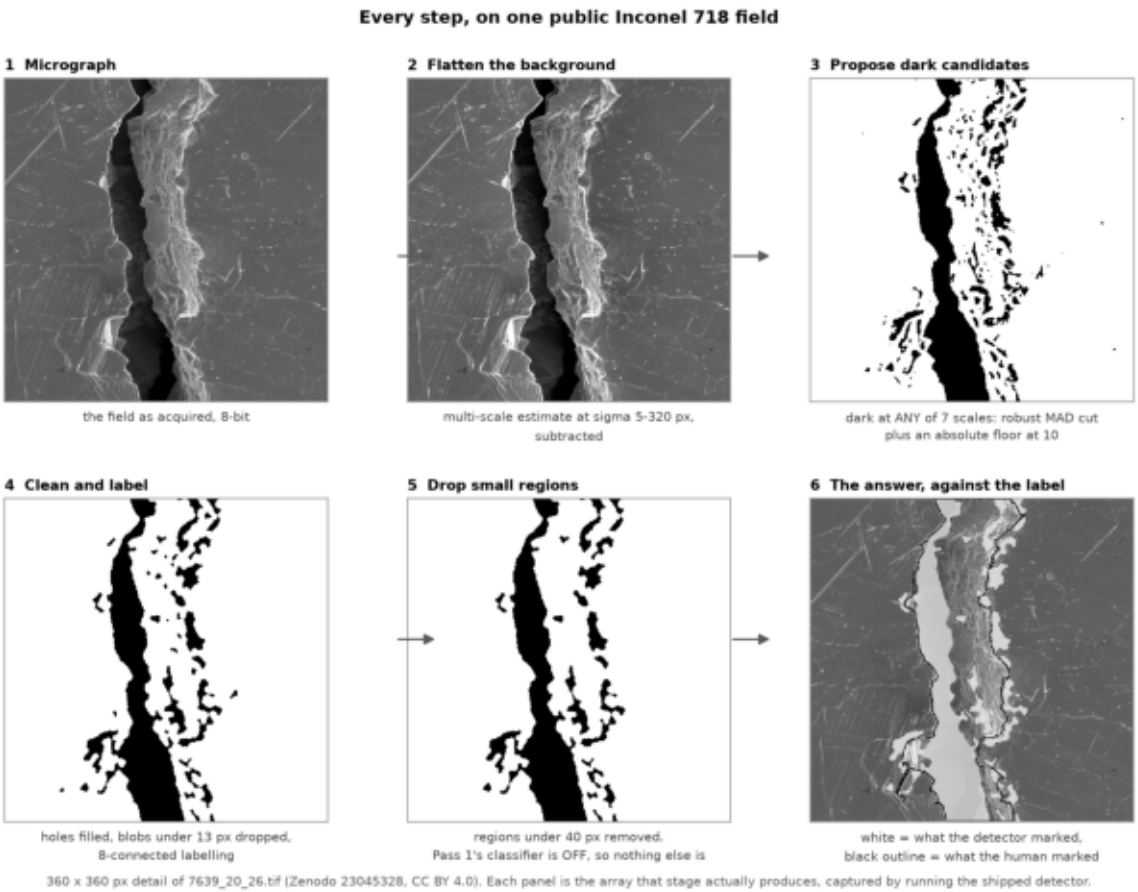


Figure 1. P6 - pipeline in pictures

Two gates do the detection: a robust median-absolute-deviation cut at mad_k standard deviations, and an absolute floor on intensity. Everything else is preparation and cleanup. There is no training step and no model file, and every panel in Figure 1 is the array that stage actually produces — captured by running the shipped code, not drawn.

Two decisions are worth stating because both were made by measurement and both went against the intuition that produced the code in the first place.

The region classifier is switched off. An eight-feature classifier scored each candidate region, and deleting it improved every metric at no cost in free parameters. Against the same candidate stage with the classifier bypassed and nothing else changed — one variable, so the difference is the classifier's own — paired median IoU +0.1753 over thirteen specimens, 11 of 13, Wilcoxon $p = 0.0061$. The shipped path also widened the flattening ladder from one radius to seven; the two changes together are +0.2296, $p = 0.0024$, and that larger figure is not the classifier's effect and must not be quoted as it. Three specimens that had scored recall 0.0000 at every candidate setting ever tried were never invisible to the proposal stage; the classifier was rejecting them. One caveat the ablation did not cover: with the classifier off the application still keeps Pass-2 rows the model itself scored not-crack, so "switched off" is a broader change in the app than in the experiment.

Tiles cannot stand in for frames. Six region-level precision rules validated on 512 px crops were rejected on whole frames: four delete 54–75% of hand-marked crack and two change no score. The widest crack in the training crops was 43.7 px; on full frames crack widths reach 178 px, so a region rule fitted on crops is extrapolating on the object it most needs to keep.

3 On public SEM data with pixel-exact ground truth

One set: the Inconel 718 crack class of a public cross-material microstructure collection (Zenodo 23045328, CC BY 4.0), 31 whole 1024 px fields of a nickel superalloy at 1.80% crack prevalence. It is the only public crack segmentation data we found in this modality and material class; Section 7 records what else was searched and rejected.

Scoring is at 0 px tolerance with mad_k chosen leave-one-image-out. Two properties of the data had to be established before any score could be trusted: the masks are anti-aliased binary rather than strictly two-valued, so the scorer checks bimodality and refuses a mask that fails it; and the images are 16-bit TIFFs, where an ordinary 8-bit conversion drove recall to 0.0012 before it was caught. Taking the high byte reproduces the dataset authors' own 8-bit PNGs exactly, so that is the conversion used.

We report whole fields rather than the collection's released crops. Every one of the 18 source images behind those crops also appears among the originals, so the two are not independent sets, and the crops are cut around their cracks — which raises prevalence from 1.80% to 4.53% and the score with it, with no change to the detector.

3.1 What it scores

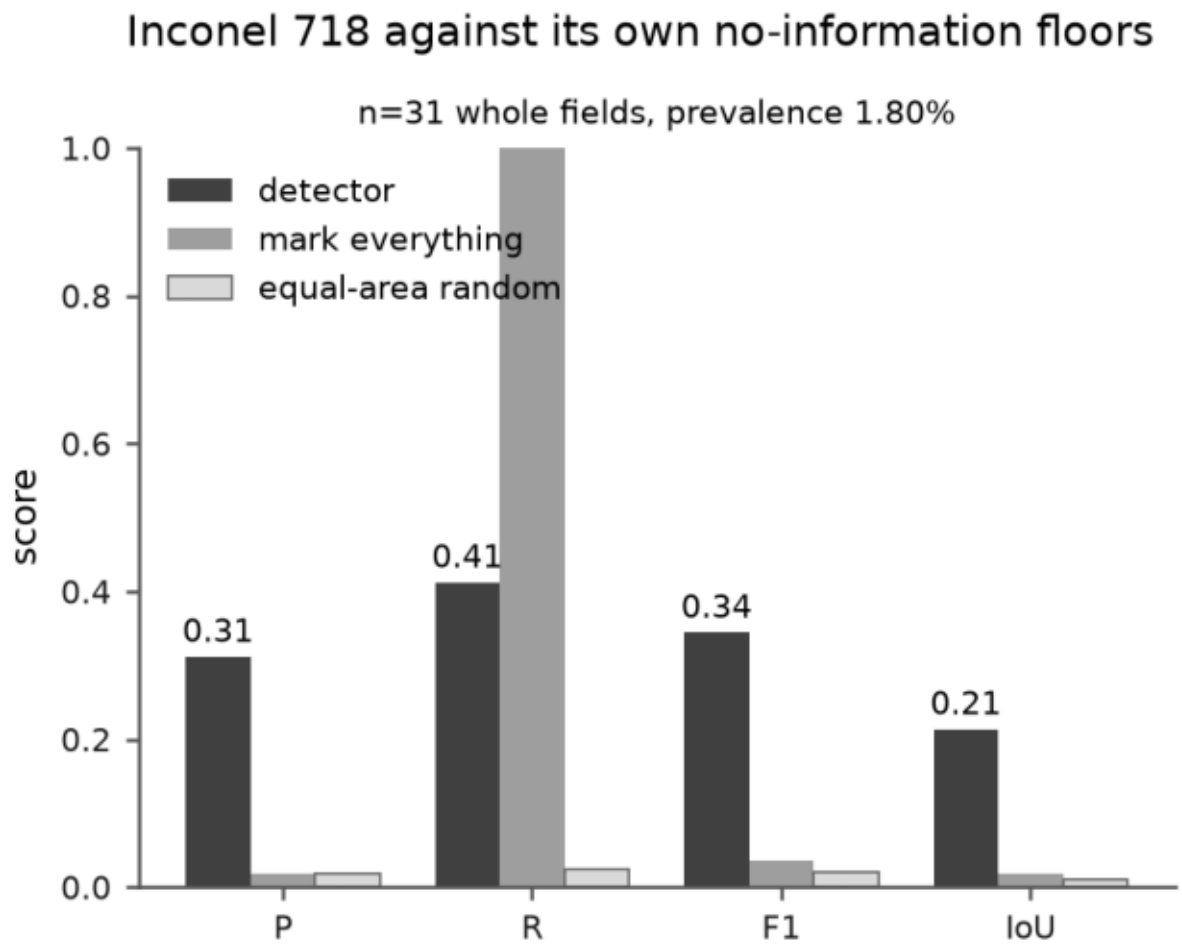


Figure 2. P1 - public scores

F1 0.3444 and IoU 0.2137, against 9.8 times the F1 and 11.8 times the IoU of marking every pixel. A mask of the same area with its pixels scattered at random scores F1 0.0196.

3.2 The one free constant, and where the sweep stopped

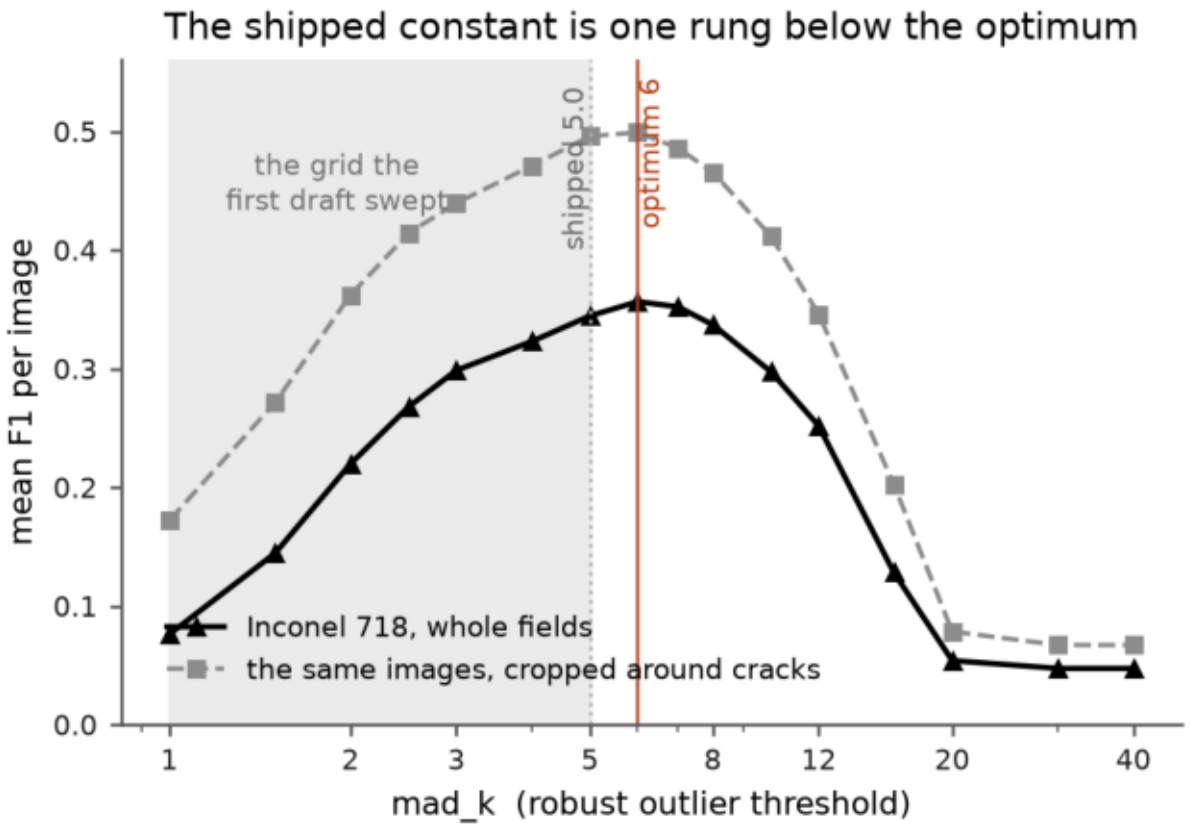


Figure 3. P2 - operating curve

mad_k = 5.0 was fitted on SEM steel in a different laboratory, on a different alloy, years before this dataset existed. An earlier draft of this paper reported that it was the best of the seven values tested here and that the curve fell away on either side of it, and offered that as the paper's strongest result.

It was the largest value tested. The seven rungs ran from 1.0 to 5.0, F1 was still climbing at the top of them, and nothing above the shipped constant had been scored. A claim that a constant lands on an independent laboratory's optimum cannot be made from a grid whose maximum is that constant, and we made it.

Extending the sweep to 40.0 gives the two-sided curve the claim needed. The optimum is 6.0, one rung above the shipped value, and the difference survives the test this paper demands of everyone else: across the 31 whole fields, 6.0 beats 5.0 on F1 in 27 of 31 images, paired Wilcoxon $p < 0.0001$, median gain 0.0122. On the area fraction it is the larger effect — 1.65x the labelled value at 5.0 against 1.27x at 6.0, better in 23 of 31 images, $p = 0.0124$ — and 1.06x at the next rung up, where F1 is 0.3521.

The direction matters more than the magnitude. Every attempt in this paper to raise the score by marking more either failed its controls or wrecked the area fraction (Section 3.5). The one change that improves both marks less.

On the released crops the same change is a wash on F1 ($p = 0.3533$) and makes the area fraction worse, 0.93x to 0.72x ($p < 0.0001$), because those crops are cut around their cracks and the detector already under-marks on them. The two framings disagree about the right operating point, which is a further reason not to read a single tuned constant as a property of the method.

What survives of the original claim is weaker and still worth stating: a constant carried unchanged from another laboratory's alloy lands one rung from the optimum, and every value

tested below it is worse. We have not changed the shipped default on the strength of 31 images of one alloy; Section 7 says what would be needed.

3.3 Scored as a pixel budget

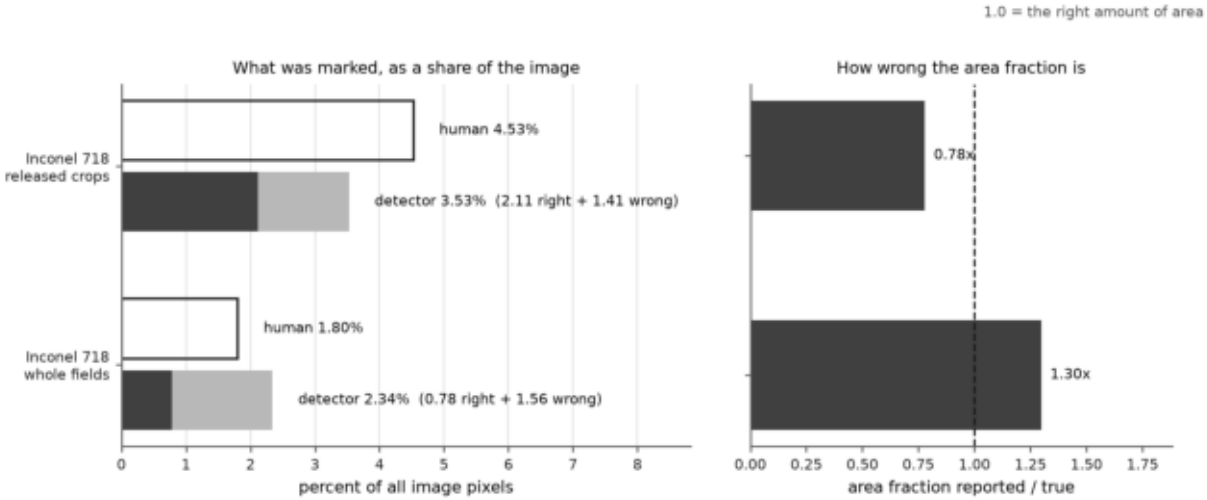


Figure 4. P5 - pixel budget

Precision, recall, F1 and IoU all discard the true negatives. On an image that is 1.80% crack that is 98% of the evidence, and it lets two very different detectors share a score — one marking 0.3% of a frame and one marking 5% can land on the same IoU. The quantity this tool reports to a user is an area fraction, a pixel percentage, so the scores belong in the same units as the answer.

	whole fields	released crops
labelled crack	1.80%	4.53%
detector marked	2.34%	3.53%
of which correct	0.78%	2.11%
of which wrong	1.56%	1.41%
area fraction reported / true	1.30x	0.78x
accuracy	97.42%	96.17%
balanced accuracy	69.82%	72.09%
MCC	0.3402	0.4840

Three things only this view shows.

The area fraction a user reads off this tool is about 1.30 times too high on whole fields. That is the error that reaches a materials conclusion, and no F1 or IoU states it.

The bias reverses between the two framings of the same images — 1.30x on whole fields, 0.78x on the crops cut from them. The detector did not change.

Accuracy is 97.42% and is worthless. Marking nothing at all would score 98.20%. It is printed here only so that it is never quoted: at this prevalence the measures worth reading are balanced accuracy and MCC, and those are 69.82% and 0.3402.

3.4 Against classical detectors

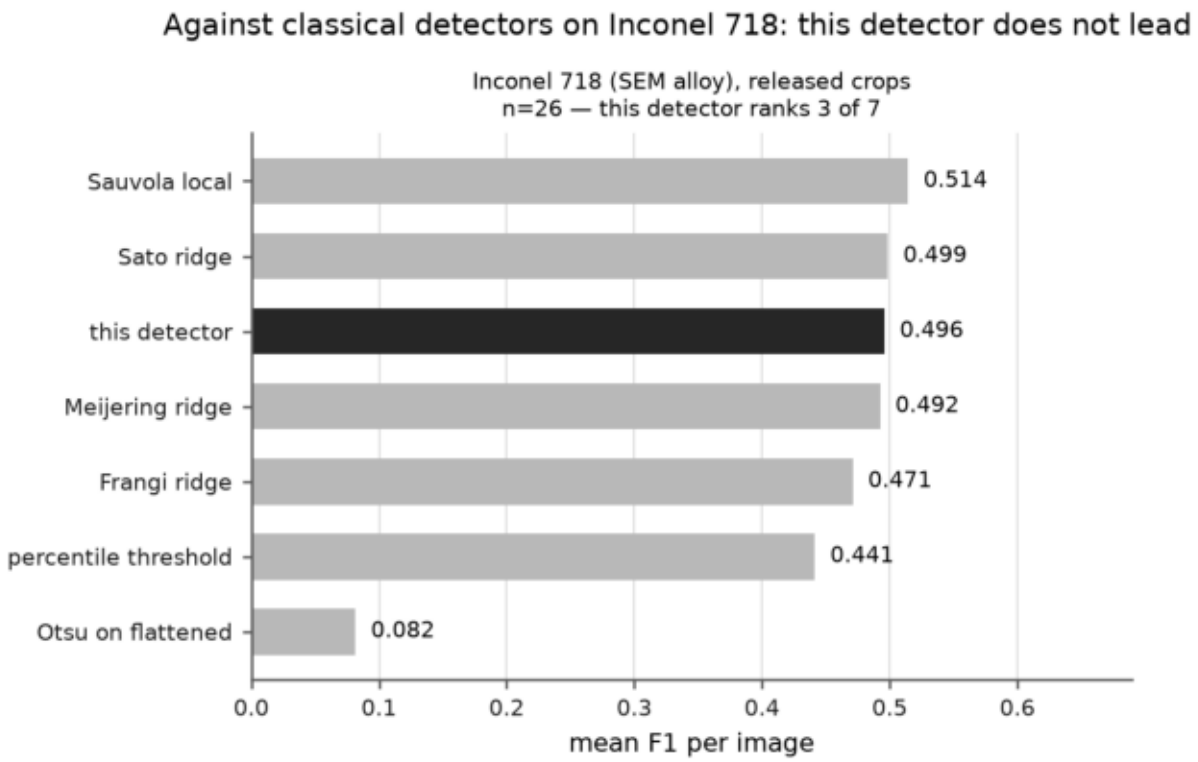


Figure 5. P4 - baselines

Every arm gets its own best parameter by the same leave-one-image-out rule, on the same images, through the same scoring function. On the released crops this detector ranks third of seven: Sauvola local thresholding reaches 0.5145 and Sato ridge 0.4985 against our 0.4958, and the top three are separated by 0.0186 F1 — about the width of a gap this project has elsewhere called irresolvable at this sample size. The honest statement there is that this detector and two classical filters are indistinguishable.

On the whole fields the rest of this paper reports, it ranks fifth of seven, at 0.3444 against the best arm's 0.4697, and fifth by a width-blind skeleton F1. That is a worse standing than the crops show, and it is the standing on the set the rest of the paper argues from. The crops flatter every arm equally, but they do not flatter them by equal amounts: the classical ridge filters lose little between the two framings and this detector loses 0.1514 F1.

We report the comparison because a reader can reproduce it in an afternoon, and because the premise is worth more than the ranking: none of these seven arms is trained, all are tuned identically, and the figure shows what the whole classical family reaches on a benchmark where the published learned state of the art is far higher.

3.5 The ranking is buyable, and we checked what it costs

We tried to raise this detector's rank. The attempt succeeded, and that is the finding.

Take the detector's own output and dilate it. Give the dilation one tuned parameter, chosen leave-one-image-out exactly as every other arm chooses its own. Nothing is added: a morphological dilation cannot find a crack the detector did not already mark, and it reads no new evidence from the image. On the released crops that mask scores 0.6307 F1 against the shipped 0.4958 — a gain of 0.1349, and first place, clear of every classical filter.

It is not an artefact of pixel F1 either. Reducing both masks to centrelines and scoring agreement within 5 px — a measure that cannot see how wide a mask is drawn — the dilated

mask still ranks first, at 0.4002 against the shipped 0.3279.
Dilation closes gaps in a fragmented mask, and gap-closing moves a centreline score.

Two further arms were run before this one and did no better. Growing the proposal through a Sato ridge response, which at least reads the image, lost to the plain dilation on the crops (0.6169 against 0.6307) and beat it by a similar margin on the whole fields (0.4730 against 0.4608) — the two framings disagree in sign, which is itself the answer. And on whole fields, growing the proposal through a different field's ridge response scored 0.4043 against the real field's 0.4065. A step whose output barely changes when you feed it the wrong image is not reading the image.

Three more channels were built and tested the same way and none survived. A local-roughness channel, motivated by what the missed crack turns out to be (Section 3.6), beat a wrong-image control by 0.0227 F1 on the whole fields and took the area fraction to 3.44x. A watershed grown from the dark core out to the gradient ridge along the crack lip — the one construction whose stopping rule is the crack edge rather than a percentile — scored below both of its own controls. Every attempt to raise the score by marking more has now failed in the same way.

So the rank can be fixed. What it costs is the two numbers this tool exists to report — both rows below are whole-field values, where the dilated mask ranks third by pixel F1 and first by skeleton F1; its first place is a released-crops result:

	area fraction, reported / true	mean width, reported / true
shipped detector	1.30x	0.75x
dilated	2.54x	2.18x

On whole fields, buying 0.1163 F1 takes the area fraction from 1.30 times the labelled value to 2.54 times it, and the mean width from 0.75x to 2.18x. A reader of this tool would be told that 2.54 times as much of the field had cracked as the annotator marked. The benchmark and the measurement move in opposite directions, so we did not ship it.

We think this is the most useful thing in this section. The ranking on this dataset is substantially a ranking of how wide each method draws, which is the same conclusion Section 4 reaches from the other direction, and it means a table of F1 scores on broad-brush labels should not be read as a table of detection ability.

3.6 Where the missing recall actually is

The shipped recall on whole fields is 0.4347 pooled over pixels — this one subsection pools, because the question is which stage loses which pixels, and a per-image mean cannot be decomposed that way; the per-image figure quoted elsewhere in this paper is 0.4123, and the two must not be mixed. The missing recall is not behind a threshold. Open both gates of the proposal stage to their limits — the MAD cut down to 0.5 and the absolute darkness floor to zero, which between them mark 53.4% of the image at precision 0.0196 — and recall still reaches only 0.579.
42.1% of the labelled crack is not dark at any scale, so no setting of either gate can propose it. That is a ceiling on this proposal mechanism for this material, not an operating point.

It also refutes a fix carried over from our own corpus, where the absolute darkness floor accounted for nearly all of the proposal-stage recall loss. On this alloy that gate costs 0.0006 recall. The gate that matters here is the MAD cut, and opening it is a precision trade rather than free recall: precision falls from 0.3214 to 0.0196.

4 What the scores measure

This is the part we would most want another group to check. Every number in this section is measured on the 26 released crops, not the whole fields the rest of the paper reports, because the constant-width construction was built against that split; Figure 6 draws its reference line from the same set. The whole-field version has not been run.

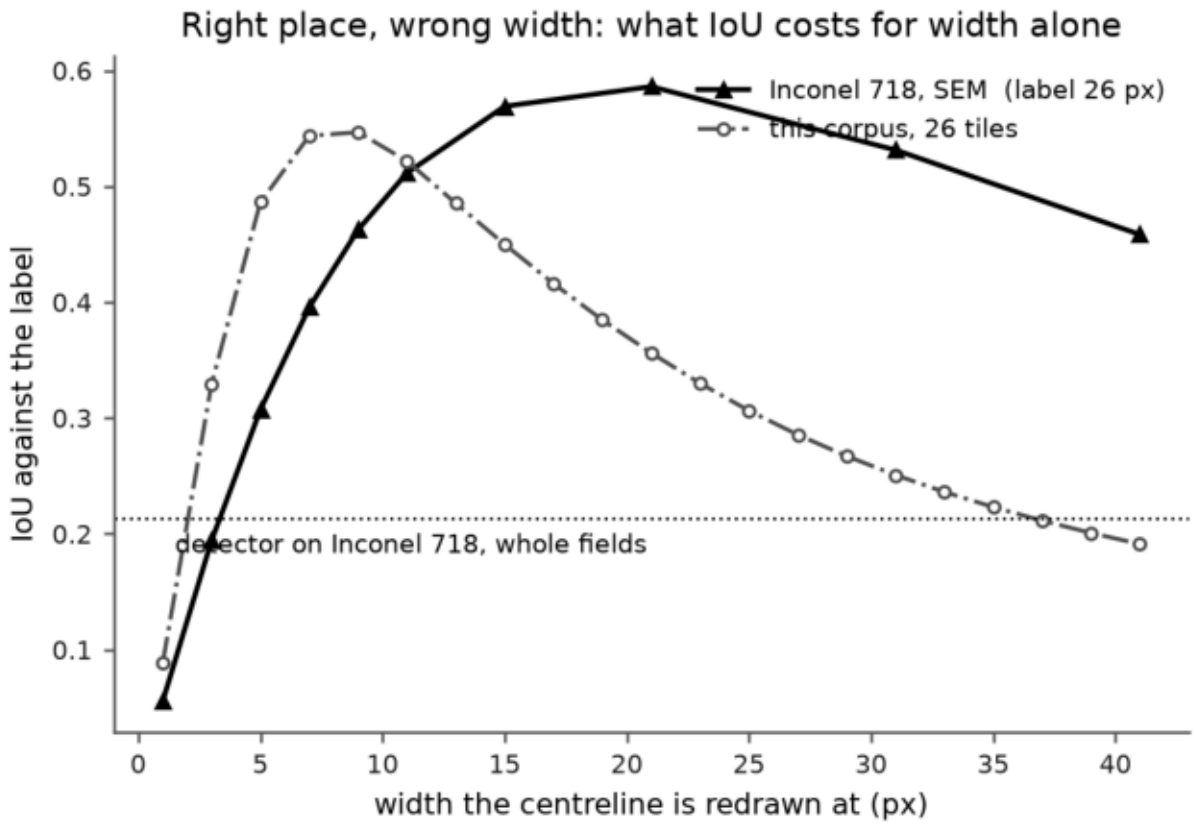


Figure 6. P3 - width sensitivity

Take the ground truth, reduce it to its centreline, and redraw it at a fixed width. The result is exactly right about where every crack is and wrong only about how wide. Scored against the label it should be nearly perfect, and it is not: the best such reconstruction reaches IoU 0.5861. Most of the available IoU is being spent on agreeing about width, and the two curves in Figure 6 peak at different widths — 21 px for this dataset's labels against 9 px for our own corpus tiles — so the same detector's IoU can differ between corpora without its localisation differing at all.

This is not an upper bound, and we presented it as one. A real label varies in width along its length, so a detector that tracks that variation can score above a constant-width redraw of the same centreline. We had to be told that by the measurement.

We also claimed width does not explain the detector's difficulty. That claim is now withdrawn as well. This dataset's labels are drawn at a median 25.8 px and its best constant-width reconstruction reaches 0.5861; our own corpus tiles, scored by the same construction, peak at 0.5467. The two ceilings are comparable. What we cannot say is that the two brush widths differ substantially, because the corpus stroke width is not measured here — Section 7 records it as unmeasured — so the inference that drew the two facts together had nothing holding it up.

That argument was stronger one draft ago. It rested on two SEM sets annotated four times apart in stroke width — 25.8 px and 6.4 px — reaching the same ceiling to within 0.007. The second of those was the EBC coating, withdrawn from this paper as a ceramic rather than a metal, and the argument is weaker without it. The measurement remains in docs/figures/sem_width_ceiling.json.

5 Negative results

These cost the most time and are the most likely to save someone else's.

Retuning the detector for a different modality does nothing. Applying the SEM-calibrated detector to transmission X-ray tomography, we swept its two gating constants singly and jointly, plus every morphology and size floor, under leave-one-specimen-out selection. No arm beat doing nothing on held-out data; the joint optimum lost 0.021 F1 when its selection was held out, and resolution-matching the scale ladder to the measured width ratio moved F1 by 0.0008.

That corpus cannot answer the question anyway. It holds four specimens, so a paired test has an exact two-sided floor of p = 0.125 — no result on it can reach significance at any conventional level, a win or a null. One specimen carries 57.7% of all adjudicated not-crack pixels, so the pooled score surface is essentially that specimen's.

F1 is width-gameable where the labels are broad. On the same corpus a by-construction-correct 3 px crack scores F1 0.1260, below marking every pixel (0.1499), while F1 climbs to 0.64 on marked width alone. "Beats the no-information floor on F1" is necessary and not sufficient; precision lift separates where F1 does not.

Six region-level precision rules validated on 512 px crops were rejected on whole frames. Four of them delete 54–75% of hand-marked crack; the other two change no score. See §2.

6 Measurement practices this forced

Stated as rules because each one followed a specific error of ours.

- Report the score in the units of the answer. F1 and IoU discard every pixel correctly left alone — 98% of the evidence on a 1.80%-crack image — and neither can state the error a user suffers, which is an area fraction off by some factor. Report labelled %, marked %, and their ratio; keep F1 and IoU beside them for comparability with the literature. The ratio is 1.30x here and 76.65x on a set this detector fails, and no F1 in this paper distinguishes those two situations as plainly.
- Never quote plain accuracy at low prevalence. It is 97.42% here, and marking nothing at all scores 98.20%. Balanced accuracy and MCC use the true negatives without being swamped by them.
- Report the no-information floor beside every score. Marking every pixel scores F1 0.0351 on this dataset. A bare score cannot be read without it, and across the sets measured for this project the floor moves by more than a factor of seven.
- Choose parameters by grouped hold-out, and say what the group is. Regrouping alone moved an AUC by +0.0458 in our data, which is larger than most effects we chased.
- A difference of medians is not a paired difference. One result moved 0.0742 as medians and 0.0000 paired.
- State the resolution of the comparison. Our laboratory corpus detects about +0.14 F1 for a new method; we reported smaller differences for weeks before measuring this.
- Controls must be able to fail. Every control we built was broken on purpose to check it noticed; two did not and were rewritten.
- Check the mask encoding and the bit depth before scoring anything. This set's masks are anti-aliased rather than two-valued and its images are 16-bit; an ordinary 8-bit conversion drove recall to 0.0012 and looked like a detector failure rather than a loading bug.
- Score whole fields, not crops of them. Cropping these images around their own cracks raises F1 by 0.1514 with no change to the method.
- An oracle must match the arity of the thing it bounds. The constant-width reconstruction in §4 is not an upper bound, and we described it as one until the data said otherwise.

7 What this draft is missing

Listed rather than hidden.

- The evidence is 31 images of one alloy, from a single released collection whose record is one sentence long and names no pixel size, no microscope and no annotation protocol. One crack class out of eighteen targets. That is enough to place the detector and not enough to characterise it, and it is the single largest weakness of this draft.
- A second class from the same collection was measured and removed. The EBC environmental barrier coating (24 fields, 0.26% prevalence) is a ceramic coating rather than a metal and is outside the application. The detector failed on it — 0.0102 precision at 0.6918 recall — and that failure is not reported above. Removing it also cost Section 4 its strongest form, which compared two sets annotated four times apart in brush width. Both measurements remain in docs/figures/.
- What was searched for and not found. Public crack segmentation data in this modality is close to non-existent. Ruled out, each for a stated reason: road and pavement sets (CrackForest, DeepCrack, Crack500, CrackSeg9k and the aggregated collections) are the wrong modality; MetalDAM is SEM steel with pixel masks but its five classes are phases and a generic "defect", not cracks; NASA's MicroNet superalloy tasks label precipitates, not cracks; a released Ti-6Al-4V fractography set (SE and BSE, 16 GB) ships images with no masks at all; and the published nickel-superalloy fatigue-crack segmentation work we found is tomography with no data release. A second independent SEM crack set would change this paper more than any further work on the method.
- Pavement results were measured and removed. The detector reaches F1 0.7932 and IoU 0.6809 on DeepCrack, ranking first of seven there. It is not reported above because a road-surface benchmark says nothing about micrographs, and the contrast between that ranking and the third-of-seven on SEM is better read as a warning about cross-modality benchmarking than as a result. The numbers remain in docs/figures/public_metrics.json.
- No laboratory results are reported. By choice: the specimen measurements are not being published here. Sections 2 and 5 cite effects measured on that corpus as methodological evidence; a reviewer cannot check those without the data, and they should be read as weaker than the public-dataset numbers.
- The corpus series in Figure 6 carries no stroke width. The instrument that would measure it comparably needs a tile cache not present on this machine.
- The detector's best score in this project is not on cracks. On MetalDAM's voids — compact, dark, high-contrast objects in SEM steel — it reaches MCC 0.5868 against 0.3402 on cracks in the superalloy, and reports an area fraction 0.74x the truth against 1.30x. That is what a multi-scale darkness threshold should do, and it bounds how much of this paper is about cracks specifically rather than about dark thin objects. Appendix A gives the full span.
- No timing, and no comparison against a trained network on equal footing. A fair speed claim needs matched hardware, and an earlier attempt found this detector 1.6x slower than a single-pass U-Net at p = 0.0001.
- Novelty is not claimed. Fifty-five framings of this work were checked against prior art and all were already owned. What is offered is measurement.

Data and code availability

The images and masks in Section 3 are the Inconel 718 crack class of a public collection, Zenodo record 23045328, CC BY 4.0 (doi:10.5281/zenodo.23045328). The whole fields are its image/mask directories and the released crops its image_cropped/mask_cropped directories; both are used unmodified except for the 16-bit to 8-bit conversion described in Section 3, which takes the high byte.

Every measurement file this paper draws a number from is listed in docs/INDEX.md with the script that produced it, and the paper itself is built by paper/make_paper.py, which fails rather than renders if a number has no source. The repository is not yet public and no archival DOI or commit hash is quoted here; both are required before submission, and

until they exist a reader cannot check any of the above. The hand-corrected micrographs of our own laboratory referred to in Sections 2, 5 and 6 are not in a public release either; docs/DATASET_RELEASE_SCOPE.md measures what releasing them would cost and lists the decisions still open.

References

Implementations of the five classical arms are those in scikit-image [10]; each was run through this paper's own scoring code, not its authors'.

[1] Otsu, N. (1979). A threshold selection method from gray-level histograms. IEEE Transactions on Systems, Man, and Cybernetics 9(1), 62–66.

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[3] Frangi, A.F., Niessen, W.J., Vincken, K.L. and Viergever, M.A. (1998). Multiscale vessel enhancement filtering. MICCAI 1998, LNCS 1496, 130–137.

[4] Sato, Y., Nakajima, S., Shiraga, N., Atsumi, H., Yoshida, S., Koller, T., Gerig, G. and Kikinis, R. (1998). Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. Medical Image Analysis 2(2), 143–168.

[5] Meijering, E., Jacob, M., Sarria, J.-C.F., Steiner, P., Hirling, H. and Unser, M. (2004). Design and validation of a tool for neurite tracing and analysis in fluorescence microscopy images. Cytometry Part A 58(2), 167–176.

[6] Olofsson, P., Foody, G.M., Herold, M., Stehman, S.V., Woodcock, C.E. and Wulder, M.A. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment 148, 42–57. doi:10.1016/j.rse.2014.02.015. The closest prior art to Section 3.3. Reporting the error matrix in units of area proportion, and correcting a map-derived area for classifier bias, is standard practice in land-cover accuracy assessment and has been for a decade. What Section 3.3 adds is not the idea but the application to a micrograph area fraction, and a full budget rather than a single bias-adjusted point estimate.

[7] Durmaz, A.R., Müller, M., Lei, B., Thomas, A., Britz, D., Holm, E.A., Eberl, C., Mücklich, F. and Gumbsch, P. (2021). A deep learning approach for complex microstructure inference. Nature Communications 12, 6272. Precedent in this modality for reporting the downstream morphological quantity beside the segmentation score, and for the observation that phase-boundary localisation limits per-object morphology.

[8] ASTM E562-19. *Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count.* ASTM International. The units a materials laboratory reports in, and the accuracy target Section 6 measures against.

[9] Underwood, E.E. and Banerji, K. Quantitative fractography. doi:10.1007/BF02698249. The roughness and profile parameters the measurement side of this tool reports.

[10] van der Walt, S., Schönberger, J.L., Nunez-Iglesias, J., Boulogne, F., Warner, J.D., Yager, N., Gouillart, E. and Yu, T. (2014). scikit-image: image processing in Python. PeerJ 2, e453.

[11] Salvato, D. et al. (2023). Impact of SEM acquisition parameters on the porosity analysis of irradiated U-Mo fuel. Nuclear Materials and Energy. doi:10.1016/j.nme.2023.101494. Published precedent that a segmentation-derived extent metric moves with acquisition settings, which is the effect Section 5 reports for detection mode.

[12] Dahari, A. et al. (2025). Prediction of microstructural representativity from a single image. Advanced Science. doi:10.1002/advs.202414149, arXiv:2410.19568. Bounds the

within-image representativity of a phase fraction analytically; it does not estimate between-patch variance over a specimen surface, which is the quantity Section 7 says this corpus cannot supply.

[13] The Inconel 718 crack class used throughout Section 3 is from a public cross-material microstructure collection, Zenodo record 23045328, CC BY 4.0. doi:10.5281/zenodo.23045328.

This list is incomplete. It covers the arms this paper runs and the prior art it has checked itself against. It does not yet contain a survey of learned crack segmentation, and the sentence in Section 3.4 about the published learned state of the art is uncited and should either be sourced or deleted before submission.

Appendix A Every set measured, as a pixel budget

All five SEM sets this project has scored, in the units the tool reports. Two of them are not crack benchmarks and say so. Percentages are of all image pixels, averaged per image, mad_k chosen leave-one-image-out.

set	n	status	lab%	mark%	ok%	bad%	rep/true	bal.acc	MCC
Inconel 718, fields	31	the paper's benchmark	1.80%	2.34%	0.78%	1.56%	1.30x	69.82%	0.3402
Inconel 718, crops	26	same images, cut on cracks	4.53%	3.53%	2.11%	1.41%	0.78x	72.09%	0.4840
EBC coating, fields	24	withdrawn: ceramic	0.26%	20.29%	0.19%	20.10%	76.65x	74.52%	0.0662
EBC coating, crops	8	withdrawn	0.61%	19.70%	0.49%	19.21%	32.54x	76.72%	0.1108
MetalDAM voids	16	NOT cracks: voids	1.95%	1.44%	1.35%	0.09%	0.74x	78.03%	0.5868

The span is the point. The same detector, unchanged, reports an area fraction that is 0.74 times the truth on compact voids in steel and 76.65 times the truth on a coating at 0.26% prevalence. On the alloy it is 1.30x. Anyone quoting a single accuracy figure for this detector is quoting one point on that range.

MetalDAM is the most revealing row and it is not a crack result. Its targets are voids: compact, dark, high-contrast. The detector reaches MCC 0.5868 there against 0.3402 on cracks in the superalloy — its best score in this project is on the objects that are easiest to be dark. That is consistent with what it is: a multi-scale darkness threshold. Cracks are hard for it because they are thin and low in contrast, not because they are cracks.

The two pavement sets are absent. They were measured on P/R/F1/IoU before being removed from this paper as the wrong modality, and their downloads were not retained, so no budget was computed for them.

Appendix B. The detector, end to end



Figure 7. The shipped pipeline. Five constants are read from analysis/detector_overrides.py when the plate is drawn; mad_k and the two size floors are typed into the generator and are not yet under that check. Pass 2 and the hand-correction layer are stages of the application and are not in the scored path.